ICT Exam Examination Access Density

Summary

The ICT Exam Access Density is a Application designed to measure the relationship between computer lab usage and ICT examination outcomes in Tanzanian government secondary schools. This system introduces the Exam Access Density (EAD) metric to evaluate lab hours as a resource distributed per exam candidate. The 2025 NECTA cycle analyzed 38 matched schools across 12 regions. These schools account for 1,404 Form 2 candidates and 437 Form 4 candidates, managed by 8 field officers. A critical finding is that some schools with thousands of total lab hours have near-zero ICT EAD scores, exposing a significant equity gap meaning that they have not conducted the ICT exams. For example, Engarenaibor SS recorded an ICT EAD of 0.0 with only 0.35 total lab hours for 21 candidates. Nkoasenga SS has the lowest confirmed ICT EAD at 3.6 hours per candidate.

The ICT EAD Framework

The ICT EAD formula divides Actual Lab Usage Hours by Total ICT Exam Candidates. It penalizes schools that dilute lab hours across large student cohorts without proportional scheduling.

The system categorizes ICT EAD scores into five tiers:

- Critical (Red): Less than 10 hours per candidate. Requires immediate hardware or scheduling intervention.
- Low (Orange): 10 to 50 hours per candidate. Indicates under-served candidates.
- Medium (Yellow): 50 to 200 hours per candidate. Represents moderate access.
- Good (Green): 200 to 500 hours per candidate. Represents healthy access density.
- High (Blue): More than 500 hours per candidate. Indicates abundant access.

Data Architecture

The application is a fully portable application requiring no server dependency or external database calls. It embeds all data in a 17KB JSON file and uses HTML5, Tailwind CSS, Chart.js, and Vanilla JavaScript.

The tool relies on two primary data files:

- Usage Log: Contains annual lab usage hours from 2019 to 2025 for 120 active schools.
- NECTA Results: Contains ICT examination results from 2018 to 2025, including grade distributions and total candidates sat.

Seven schools in the 2025 matched set lack current-year usage data and cannot receive ICT EAD scores.

Analytical Tools

Tab 1: ICT EAD Scorecard

The Scorecard is a ranked and filterable reference table detailing ICT EAD scores, pass rates, grade distributions, and diagnostic flags for every matched school. Users can filter by school name, form level, field officer, region, and ICT EAD tier. Amani Abeid Karume SS represents an extreme outlier in this view, boasting the highest dataset ICT EAD of 4,793.3 but a 0% pass rate.

Tab 2: Bottleneck Triage

This tab converts rankings into an actionable dispatch queue. Schools receive Priority Levels ranging from P1 to P4:

- P1 Critical (Red): Triggered by an ICT EAD below 10, or an ICT EAD below 20 combined with a pass rate under 30%. Recommends immediate hardware audits or scheduling interventions.
- P2 High Risk (Orange): Triggered by an ICT EAD below 20 or a fail rate over 50%. Recommends schedule reviews and exam prep pushes.
- P3 Watch (Amber): Triggered by missing usage data or a pass rate under 40%. Recommends field data collection.

- P4 Monitor (Green): Applied to schools with normal performance and no active flags. Recommends routine checks.

Tab 3: Candidate Load vs. Hours Correlation

This scatter chart maps all schools simultaneously to reveal systemic patterns. The X-axis represents ICT exam candidates, and the Y-axis represents actual lab hours. Bubble size reflects the ICT EAD score, while bubble color indicates the pass rate tier. This visualizes scaling challenges, systemic risks, under-utilization, and benchmark schools.

Field Team Members Portfolios

The 38 matched Form 2 schools are divided among 8 field officers. Danny's portfolio holds the highest average ICT EAD at 401.0 hours per candidate, primarily driven by strong schools like Einoti SS and Hawa Mchopa SS.

Officer	Schools Managed	Avg EAD (h/c)	Avg Pass Rate
Erick	9	31.8	72.5%
Jastin	9	88.2	37.4%
Danny	7	401.0	29.7%
Abubakar	6	63.5	54.7%
James	3	82.5	28.6%
Janeth	2	46.7	86.4%
Ashura	2	32.7	39.2%
Juma	3	1,614.0	41.4%

Limitations & Next Steps

ICT EAD calculations represent the maximum possible hours available per candidate, operating under the assumption that all lab hours strictly serve exam students. The formula utilizes actual exam participants rather than total classroom enrollment to ensure accuracy.

Immediate Actions:

- Dispatch hardware audits to P1 critical schools, including Nkoasenga SS, Mwapachu SS, and Engarenaibor SS.
- Resolve the missing 2025 usage data for the 7 unrecorded schools.

- Investigate Amani Abeid Karume SS regarding its anomalous ICT EAD and 0% pass rate.

Long-Term Recommendations:

- Implement a formal collection process for ICT exam candidate counts to forecast ICT EAD.
- Integrate ICT EAD as a fifth health dimension into the existing Academic Leverage Index (ALI) dashboard.
- Add systemic enhancements like a PDF export function for the Triage table and historical ICT EAD trend charts.

Data Source

- https://necta.go.tz